

Yaojia Han

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Fields

Primary: Computational Mechanics, Fatigue Life Prediction, Uncertainty Quantification, Machine Learning for Engineering, Finite Element Analysis

Secondary: Composite Materials, Progressive Damage Simulation, Reliability Analysis, Multi-body Dynamics

Education

Fudan University, China

Doctor of Engineering, Electronic Information

Sep 2022 – Jun 2026

Dissertation: *Performance Degradation Mechanisms and Multiaxial Fatigue Life Prediction Methods of Civil Aircraft Door Mechanism Systems*

Northwestern Polytechnical University, China

Master of Aeronautical Engineering

Sep 2019 – Mar 2022

Dissertation: *Design and Analysis of Reliability Test of an Electro-pneumatic Valve*

Bachelor of Safety Engineering

Sep 2014 – Jun 2018

Dissertation: *Qualitative Reliability Analysis of an Aircraft Cabin Door Lock System*

Publications

Journal Articles

Han, Y.-J., Li, C.-F., & Fei, C.-W. (2026). Two-Scale Continuum Damage Mechanics Model with Separate Deviatoric and Volumetric Damage for Multiaxial Fatigue Life Prediction. *International Journal of Fatigue* [\[Link\]](#)

Han, Y.-J., Sun, Y.-P., & Fei, C.-W. (2026). CDM–Bayesian framework for multiscale fatigue life prediction of aircraft mechanisms with rigid–flexible coupling. *Aerospace Science and Technology* [\[Link\]](#)

Han, Y.-J., Zheng, B.-Y., & Fei, C.-W. (2026). Multi-source reliability evaluation method of aircraft mechanism performance considering wear-plasticity accumulative damage. *Engineering Failure Analysis* [\[Link\]](#)

Fei, C.-W., **Han, Y.-J.**, Wen, J.-R., Li, C., Han, L., & Choy, Y.-S. (2024). Deep learning-based modeling method for probabilistic LCF life prediction of turbine blisk. *Propulsion and Power Research* [\[Link\]](#)

Review Articles

Fei, C.-W., Li, C., Lin, J.-Y., **Han, Y.-J.**, Choy, Y.-S., & Chen, C.-H. (2024). Structural design of aeroengine radiators: State of the art and perspectives. *Propulsion and Power Research* [\[Link\]](#)

Conference Papers

Han, Y.-J., Zhao, Y., He, D., Li, X., Jiang, D., & Cui, W. (2022). Influence of temperature on fatigue life of lock test mechanism. *QR2MSE*, pp. 1599–1606. [\[Link\]](#)

Han, Y.-J., Jia, J., Cui, W., Wan, F., & Zhang, Y. (2021). A Neural Network Based Approach for Reliability Analysis of UAV Flapping Wing. *IEEE SDPC*, pp. 42–47. [\[Link\]](#)

Han, Y.-J., Cui, W., Wan, F., Jiang, D., Li, X., & He, D. (2021). Design and Dynamic Analysis of Random Vibration Test Fixture for Aerospace Valve. *PHM-Nanjing*, pp. 1–6. [\[Link\]](#)

Research Experience

Performance Degradation and Reliability of Complex Mechanisms

- Developed physics-informed degradation models capturing coupled wear and cyclic plasticity effects.
- Quantified impacts on motion accuracy and functional reliability.
- Established a multi-source uncertainty framework for predictive reliability assessment.

Multiaxial Fatigue Modeling via Two-Scale CDM

- Proposed a two-scale continuum damage mechanics framework linking material and structural scales.
- Introduced deviatoric/volumetric damage decomposition for multiaxial fatigue.
- Enabled interpretable fatigue life prediction under complex loading.

Failure Mechanisms in Additively Manufactured Structures

- Established an experimental–computational framework for defect characterization.
- Quantified defect effects on stress response and fatigue life distribution.
- Provided insights for defect-tolerant structural design.

Optimization of Gradient-Material Structures in Additive Manufacturing

- Developed a material–structure integrated optimization framework.
- Addressed transition-zone design via constitutive modeling and gradient optimization.
- Improved structural performance and reliability through multi-objective design.

Professional Activities

Presentations

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| 2026 | - 11th International Conference on Engineering Failure Analysis. (A data-driven fatigue reliability assessment of motion mechanisms based on surrogate models.) |
| 2021 | <ul style="list-style-type: none">- 2021 5th IEEE International Conference on Sensing, Diagnostics, Prognostics, and Control- 2021 Global Reliability and Prognostics and Health Management (PHM-Nanjing)- 2021 The 11th International Conference on Quality, Reliability, Risk, Maintenance, and Safety Engineering |

Honors and Awards

Outstanding Graduate , Fudan University, China	2026
Best Paper Award , Shanghai Aeronautics Academy, China	2025
Outstanding Doctoral Student Academic Scholarship , Fudan University, China	2023
Best Paper Award , Shanghai Aeronautics Academy, China	2022

Patents

Fei, C.W., **Han, Y.-J.**, “A Method and System for Predicting the Low-Period Fatigue Life of a Turbine Blade Disk,” Chinese Patent CN202211283229.1, 2022. (In Chinese)

Technical Skills

Finite element analysis: ANSYS, Abaqus

Multibody dynamics modeling: Adams, RecurDyn

Numerical computation and data analysis: MATLAB, Python

3D modeling and structural design: CATIA, SolidWorks, Siemens NX (UG)

Languages

Mandarin (native), English (PTE:65)

Last updated: June 1 , 2026